

Aangan — Buy List

Nathaniel School of Music · house + studio Raspberry Pi system · 29 Jul 2026 · every item is a clickable link

How to use this: buy **BOTH projects now, as two SEPARATE orders** — one cart/invoice for Project 1 (Studio & House), one for Project 2 (Music rig) — so each project stays independent and trackable. Before the Silverline trip, WhatsApp **+91 70909 39819** and confirm stock (139/5 VT Complex, opp. Karthik Plaza, Nagarathpete, SP Road, Bangalore). If a search link shows several options, pick the exact name written here. Prices are rough estimates for budgeting only.

Already owned — do NOT buy: Raspberry Pi 5 8GB · Pianoteq Pro licence · ESP32 x6 · ESP32-CAM · MC-38 door sensors x9 · water-leak sensors x6 · WS2812B LED strip · DHT22 · PIR x4 · MQ-2 · INMP441 · jumper wires/PCBs/enclosures · acoustic weatherstrip · old iPad · certified smoke + LPG alarms · **USB A→B keyboard cable** · **XLR cables** · **USB-C 100W cables** · **USB SSD** · **gaffer tape + velcro ties** · **studio UPS/power backup (no APC needed)**.

PROJECT 1 — Studio & House Pi ORDER NOW · one Silverline visit + one robu.in cart + one Amazon order

A · SILVERLINE ELECTRONICS (SP ROAD / SILVERLINEELECTRONICS.IN)

ITEM	QTY	EST.	LINK
Raspberry Pi 5 — 8GB the House brain	1	₹8,000	silverline — Pi 5 8GB
Official Raspberry Pi 27W USB-C Power Supply never a phone charger	1	₹1,300	silverline — search 27W PSU
Raspberry Pi Zero 2 W drives the two door screens	2	₹1,800 ea	silverline — search Zero 2 W · robu fallback
Official Raspberry Pi Zero 2 W PSU (or any good 5V/2.5A micro-USB)	2	₹500 ea	silverline — search PSU

B · ROBU.IN — PUT EVERYTHING IN ONE CART

ITEM	QTY	EST.	LINK
Waveshare 10.1" HD Capacitive Touch LCD (G), 1920x1200, toughened glass the two door screens — must be the "(G)" model	2	₹14,000 ea	robu — Waveshare 10.1 (G)
Micro-HDMI to HDMI cable, 1.5m Pi Zero → screen (Zero uses <i>mini</i> -HDMI — buy 2x mini-HDMI to HDMI)	2	₹250 ea	robu — mini-HDMI cable
MQ-6 LPG gas sensor module one per stove point — MQ-6, not MQ-2	4	₹120 ea	robu — MQ-6

Water leak / rain drop sensor module adds to the 6 owned → covers every sink + stove line	6	₹80 ea	robu — water sensor
DFRobot Gravity SEN0232 sound level meter the real dBA meter — studio + teaching room	2	₹3,500 ea	robu — SEN0232
HLK-LD2410 mmWave presence sensor	1	₹350	robu — LD2410
SW-420 vibration sensor module	5	₹70 ea	robu — SW-420
IR flame sensor module	4	₹70 ea	robu — flame sensor
JSN-SR04T waterproof ultrasonic sensor water tanks — sump + overhead	2	₹450 ea	robu — JSN-SR04T
HX711 + 50kg load cells (set of 4) LPG cylinder weight scale	1 set	₹400	robu — HX711 set
SanDisk Extreme A2 microSD 64GB 1 for House Pi + 2 spares for panels	3	₹700 ea	robu — SanDisk 64GB

NOT intern items (the electrician sources these): latching panic switches ×4, NC loop cable, battery-backed sounder, volt-free relay modules, motor contactor + float switches. A **low-voltage / fire-alarm technician** does this wiring — booked separately.

PROJECT 2 — Music Pi (the piano rig) ORDER NOW · separate cart & invoice · uses the Pi 5 already owned

A · THE AUDIO CORE

ITEM	QTY	EST.	LINK
Raspberry Pi DAC Pro (IQAUDIO DAC Pro — PCM5242) THE sound board · powered by the Pi, no soldering	1	₹2,500	robu — DAC Pro · search
XLR Board for IQAUDIO DAC Pro gives balanced XLR sockets · ask robu first if they can order it in ; otherwise import	1	₹2,000 + ship	robu — ask/search · pishop.us — import
Fallback if XLR board unavailable: Behringer DI400P passive stereo DI ×2 (or one DI20) DAC Pro RCA → DI → balanced XLR. Only buy if the XLR board can't be sourced.	—	₹2,000	amazon — DI400P
6-pin female header (2.54mm) only needed if the DAC Pro arrives with bare P7/P9 pads — ask the seller which revision	2	₹30	robu — header

B · CONTROL, COOLING, POWER

ITEM	QTY	EST.	LINK
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Raspberry Pi Touch Display 2 (7") on-rig screen for switching pianos	1	₹5,500	silverline — Touch Display 2
Official Raspberry Pi 5 Active Cooler fit BEFORE the DAC goes on	1	₹500	silverline — Active Cooler · robu
Official 27W USB-C PSU	1	₹1,300	silverline — 27W PSU
100W USB-C PD power bank (e.g. Anker 737 / 20,000mAh+ with 100W PD) stage power with no wall socket	1	₹10,000	amazon — 100W PD bank
SanDisk Extreme A2 microSD 64GB one runs the rig, one is the cloned spare in the case	2	₹700 ea	amazon — SanDisk 64GB

C · TRANSPORT

ITEM	QTY	EST.	LINK
Hard foam-lined case △ DO NOT buy yet — the stack must be measured first (the XLR board sits above the Pi's ports; normal Pi cases don't close). Cables (keyboard USB, XLR, USB-C) are already owned.	1	₹3,000	buy after assembly

Aangan · Nathaniel School of Music · prepared 29 Jul 2026 · Project 1 ≈ ₹50k (the two screens are most of it) · Project 2 ≈ ₹23–25k · Full step-by-step builds: “Aangan — Install Guide” PDF (house) and PIANO_PI_BUILD_PLAN.html (music) in the aangan repo.